NEXT STEPS: EXECUTION ROADMAP

Hybrid Modeling of Fluoride Adsorption Using Coconut Husk

Date: May 3, 2026
Status: Phase 1 Research Complete → Phase 1 Execution (Data Generation)
Objective: Generate realistic experimental dataset and move to Langmuir fitting

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IMMEDIATE WORKFLOW

```
Phase 1 (COMPLETE)      Phase 1 (NOW)           Phase 2 & Beyond
  Literature Review      Project Setup           Langmuir Fitting
  Parameter Extraction   DoE Matrix Gen         Residual Analysis
  Mechanism Study        Simulation Function    ML Training
                          Data Validation        Hybrid Model
                          Dataset Creation       Dashboard/Report
```

STEP-BY-STEP ACTION PLAN

STEP 1: Project Setup & Environment (30 min)

```
# Create project directory
mkdir fluoride-adsorption-hybrid
cd fluoride-adsorption-hybrid

# Create folder structure
mkdir -p {data,src,notebooks,models,app,reports,output}

# Verify structure
tree -L 2  # or 'ls -la' for Windows/Mac
```

1.1 Create Project Directory Structure

Expected output:

```
fluoride-adsorption-hybrid/
   data/              # Datasets and DoE matrices
```

```
src/                # Python source code
notebooks/          # Jupyter notebooks for analysis
models/             # Saved ML models
app/                # Streamlit dashboard
reports/            # Final reports and documentation
output/             # Generated figures and outputs
```

```
# Create virtual environment
python -m venv venv

# Activate it
# On macOS/Linux:
source venv/bin/activate

# On Windows (PowerShell):
venv\Scripts\Activate.ps1

# On Windows (Command Prompt):
venv\Scripts\activate.bat
```

1.2 Create Virtual Environment

```
# Core scientific stack
pip install numpy pandas scipy scikit-learn matplotlib seaborn

# DoE and modeling
pip install pyDOE2 xgboost

# Interactive notebooks
pip install jupyter jupyterlab

# Dashboard (for later)
pip install streamlit

# Save requirements for reproducibility
pip freeze > requirements.txt
```

1.3 Install Dependencies

```
# Initialize git repository
git init

# Create .gitignore
cat > .gitignore << EOF
venv/
```

```
__pycache__/
*.pyc
*.pyo
*.egg-info/
.ipynb_checkpoints/
.DS_Store
*.csv  # Optionally exclude large datasets
EOF

# First commit
git add .
git commit -m "Initial project setup: Phase 1 execution"
```

1.4 Initialize Git (Optional but Recommended) **Status:** Environment ready for data generation

---

STEP 2: Generate DoE Matrix (45 min)

2.1 Create DoE Design Script Create file: `src/doe_design.py`

This script generates a **Face-Centered Central Composite Design (CCD)** with: - **5 factors:** pH, Initial Concentration, Time, Temperature, Flow Rate - **48-54 runs:** Covers all combinations and center points - **Literature-validated ranges:** Based on Phase 1 research

Key parameters:

```
pH:                  3.0 - 9.0      (optimum 6.0-7.0)
C :                 1.0 - 10.0 mg/L (WHO contamination range)
Time:                10 - 120 min   (rapid to equilibrium)
Temperature:         20 - 40 °C     (ambient to warm)
Flow Rate:           0.5 - 2.0 L/min (column operation)
```

```
# Navigate to project directory
cd fluoride-adsorption-hybrid

# Run DoE script
python src/doe_design.py
```

2.2 Run DoE Generation **Expected output:**

```
================================================================================
CENTRAL COMPOSITE DESIGN (CCD) GENERATION
Fluoride Adsorption on Coconut Husk - 5 Factor DoE
================================================================================

[1/4] Generating face-centered CCD matrix...
      Design shape: (48, 5)
```

```
      Total experimental runs: 48

[2/4] Scaling to physical units...
      Scaling complete
    Factor ranges:
       pH        : 3.00 - 9.00 -
       C0        : 1.00 - 10.00 mg/L
       Time      : 10.00 - 120.00 min
       Temp      : 20.00 - 40.00 °C
       Flow      : 0.50 - 2.00 L/min

[3/4] Creating experimental design table...
      DataFrame created

[4/4] Saved to: data/doe_matrix.csv

================================================================================
DESIGN STATISTICS
================================================================================
Total runs:                 48
Factorial points (2^5):     32
Axial/star points (2k):     10
Center points:              6
Levels per factor:          3 (low, center, high)
Design type:                Face-Centered CCD (  = 1.0)
```

Output file: `data/doe_matrix.csv` - 48 rows (experimental runs) - 8 columns: Run, pH, C0, Time, Temp, Flow, Order, (plus metadata)

```
# Quick inspection
python -c "
import pandas as pd
df = pd.read_csv('data/doe_matrix.csv')
print('DoE Matrix Summary:')
print(df.head(10))
print(f'\nShape: {df.shape}')
print(f'\nDescriptive statistics:')
print(df[['pH', 'C0', 'Time', 'Temp', 'Flow']].describe())
"
```

2.3 Verify DoE Matrix **Status:** DoE matrix generated and validated

---

STEP 3: Build Realistic Simulation Function (1.5 hours)

3.1 Understanding the Physics-Based Model The simulation incorporates **6 key mechanisms** from literature:

1. **Langmuir Equilibrium** (Baseline physics)

```
q_e = (q_max × K_L × C_e) / (1 + K_L × C_e)
```

- q__max = 8.5 mg/g (literature consensus)
- K__L = 0.12 L/mg at 25°C

2. **Temperature Dependence** (Arrhenius-like)

```
K_L(T) = K_L,ref × exp[E_a/R × (1/T_ref - 1/T)]
```

- E__a = 20 kJ/mol (endothermic process)
- Accounts for ~5-10% capacity increase per 20°C

3. **pH Effects** (Gaussian bell curve)

```
pH_factor = exp[-(pH - 6.5)² / (2 × 1.5²)]
```

- Peak at pH 6.5 (optimal for coconut husk)
- 30-40% reduction at extremes (pH 3 or 9)

4. **Pseudo-Second-Order Kinetics** (Rate-limiting step)

```
q_t = (q_e² × k  × t) / (1 + q_e × k  × t)
```

- k = 0.05 g/(mg · min)
- Reaches ~70% equilibrium at 30 min, >95% at 120 min

5. **Flow Rate Effect** (Contact time limitation)

```
flow_factor = 1 / (1 + flow_rate)
```

- Inverse relationship: higher flow → lower removal

6. **Realistic Noise** (Analytical error)

```
noise = Normal( =0,  =±5% efficiency)
```

- ±5% standard deviation (typical lab error)

3.2 Create Simulation Script

Create file: `src/simulate_adsorption.py`

The script includes: - `FluorideAdsorptionSimulator` class with all physics - Literature-validated parameters - Batch simulation for all DoE runs - Validation checks against literature trends

```
# Run simulation
python src/simulate_adsorption.py
```

3.3 Run Simulation

Expected output:

```
================================================================================
FLUORIDE ADSORPTION SIMULATION
Physics-Based Model with Literature-Validated Mechanisms
================================================================================

[1/4] Loading DoE matrix...
```

```
    Loaded 48 experimental runs

[2/4] Initializing simulator...
    Simulator ready with literature-validated parameters:
      qmax = 8.5 mg/g
      KL = 0.12 L/mg (at 25°C)
      pH_opt = 6.5
      E_a = 20 kJ/mol
      Noise level = ±5.0%

[3/4] Simulating all runs...
    Generated data for 48 runs

[4/4] Validating simulation...

================================================================================
SIMULATION VALIDATION AGAINST LITERATURE
================================================================================

[CHECK 1] pH Dependence (should peak at pH 6-7):
  pH 3.0:  45.23% removal
  pH 5.0:  78.45% removal
  pH 6.5:  89.12% removal (PEAK)
  pH 8.0:  71.34% removal
  pH 9.0:  42.67% removal

[CHECK 2] Time Dependence (should saturate by 120 min):
  t= 10 min:  28.90% removal
  t= 30 min:  65.34% removal (70% of eq)
  t= 60 min:  82.45% removal
  t=120 min:  94.78% removal (>95% of eq)

[CHECK 3] Temperature Effect (should increase):
  T=20°C: 76.45% removal
  T=30°C: 81.23% removal
  T=40°C: 85.67% removal (  increasing)

[CHECK 4] Flow Rate Effect (should decrease with higher flow):
  Q=0.50 L/min: 88.45% removal
  Q=1.25 L/min: 75.23% removal
  Q=2.00 L/min: 62.34% removal (  decreasing)

[SUMMARY] Data Statistics:
  Efficiency range: 15.23 - 97.45 %
  Capacity range:   0.45 - 8.23 mg/g
  Mean efficiency:  71.34 %
  Std efficiency:   22.45 %
   Validation complete
```

```
 Saved to: data/dataset_simulated.csv

================================================================================
PHASE 1.2 COMPLETE: Realistic Data Generated
================================================================================

Next: Run Langmuir fitting in Phase 2
File: data/dataset_simulated.csv
```

Output file: `data/dataset_simulated.csv` - 48 rows (experimental runs) - Columns: - Input factors: pH, C0_mg_L, Time_min, Temp_C, Flow_L_min - Response variables: Efficiency_percent, Capacity_mg_g - Equilibrium capacity for reference

Status: Realistic data generated with validation

STEP 4: Review Generated Data (15 min)

```
# Load and inspect data
python << 'EOF'
import pandas as pd
import numpy as np

# Load dataset
df = pd.read_csv('data/dataset_simulated.csv')

print("="*80)
print("GENERATED DATASET SUMMARY")
print("="*80)
print(f"\nDataset shape: {df.shape}")
print(f"Columns: {list(df.columns)}")

print("\n" + "-"*80)
print("FIRST 10 RUNS:")
print("-"*80)
print(df.head(10))

print("\n" + "-"*80)
print("DESCRIPTIVE STATISTICS:")
print("-"*80)
print(df[['pH', 'C0_mg_L', 'Time_min', 'Temp_C', 'Flow_L_min',
          'Efficiency_percent', 'Capacity_mg_g']].describe())

print("\n" + "-"*80)
print("EFFICIENCY DISTRIBUTION:")
print("-"*80)
```

```
print(f"Min:  {df['Efficiency_percent'].min():.2f}%")
print(f"Max:  {df['Efficiency_percent'].max():.2f}%")
print(f"Mean: {df['Efficiency_percent'].mean():.2f}%")
print(f"Std:  {df['Efficiency_percent'].std():.2f}%")

print("\n  Dataset ready for Phase 2 analysis")
EOF
```

4.1 Quick Data Summary

4.2 Data Visualization

Create file: `notebooks/01_data_exploration.py`

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# Load data
df = pd.read_csv('data/dataset_simulated.csv')

# Create figure with 6 subplots
fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Generated Dataset: Fluoride Adsorption Simulation',
             fontsize=16, fontweight='bold')

# 1. pH Effect (should show bell curve)
axes[0,0].scatter(df['pH'], df['Efficiency_percent'], alpha=0.6, s=80)
axes[0,0].axvline(x=6.5, color='r', linestyle='--', label='Optimum pH')
axes[0,0].set_title('pH Effect', fontsize=12, fontweight='bold')
axes[0,0].set_xlabel('pH')
axes[0,0].set_ylabel('Removal Efficiency (%)')
axes[0,0].grid(True, alpha=0.3)
axes[0,0].legend()

# 2. Time Effect (should show saturation)
axes[0,1].scatter(df['Time_min'], df['Efficiency_percent'], alpha=0.6, s=80)
axes[0,1].set_title('Contact Time Effect', fontsize=12, fontweight='bold')
axes[0,1].set_xlabel('Time (min)')
axes[0,1].set_ylabel('Removal Efficiency (%)')
axes[0,1].grid(True, alpha=0.3)

# 3. Temperature Effect (should be positive)
axes[0,2].scatter(df['Temp_C'], df['Efficiency_percent'], alpha=0.6, s=80)
axes[0,2].set_title('Temperature Effect', fontsize=12, fontweight='bold')
axes[0,2].set_xlabel('Temperature (°C)')
axes[0,2].set_ylabel('Removal Efficiency (%)')
axes[0,2].grid(True, alpha=0.3)

# 4. Concentration Effect
```

```python
axes[1,0].scatter(df['C0_mg_L'], df['Efficiency_percent'], alpha=0.6, s=80)
axes[1,0].set_title('Initial Concentration Effect', fontsize=12, fontweight='bold')
axes[1,0].set_xlabel('C  (mg/L)')
axes[1,0].set_ylabel('Removal Efficiency (%)')
axes[1,0].grid(True, alpha=0.3)

# 5. Flow Rate Effect (should be negative)
axes[1,1].scatter(df['Flow_L_min'], df['Efficiency_percent'], alpha=0.6, s=80)
axes[1,1].set_title('Flow Rate Effect', fontsize=12, fontweight='bold')
axes[1,1].set_xlabel('Flow Rate (L/min)')
axes[1,1].set_ylabel('Removal Efficiency (%)')
axes[1,1].grid(True, alpha=0.3)

# 6. Distribution
axes[1,2].hist(df['Efficiency_percent'], bins=15, edgecolor='black', alpha=0.7)
axes[1,2].axvline(df['Efficiency_percent'].mean(), color='r',
                   linestyle='--', linewidth=2, label=f"Mean: {df['Efficiency_percent'].mean()
axes[1,2].set_title('Efficiency Distribution', fontsize=12, fontweight='bold')
axes[1,2].set_xlabel('Removal Efficiency (%)')
axes[1,2].set_ylabel('Frequency')
axes[1,2].legend()
axes[1,2].grid(True, alpha=0.3, axis='y')

plt.tight_layout()
plt.savefig('output/01_data_exploration.png', dpi=300, bbox_inches='tight')
print("  Saved: output/01_data_exploration.png")
plt.show()
```

Run visualization:

```
cd fluoride-adsorption-hybrid
python notebooks/01_data_exploration.py
```

Status: Data reviewed and visualized

7-DAY ROADMAP

Day	Phase	Task	Estimated Time	Status
Day 1-2	Phase 1	Literature Review & Parameter Extraction	8-10 hrs	**DONE**

Day	Phase	Task	Estimated Time	Status
Day 2	Phase 1	Project Setup + DoE Generation + Simulation	3-4 hrs	**TODAY**
Day 3	Phase 1	Data Validation & Exploration	2 hrs	**TOMORROW**
Day 4	Phase 2	Langmuir Model Fitting (SSL)	3 hrs	**Next**
Day 4	Phase 2	Dual-Site Langmuir (DSL) Testing	2 hrs	**Next**
Day 5	Phase 3	Residual Analysis & Pattern Detection	3 hrs	**Next**
Day 6	Phase 4	ML Feature Engineering & Model Training	4 hrs	**Next**
Day 7	Phase 5	Hybrid Model Integration & Comparison	3 hrs	**Final**
Day 8	Phase 6	Streamlit Dashboard Development	4 hrs	**Final**
Day 9-10	Phase 7-8	Report Writing & Visualization	8 hrs	**Final**

WHAT YOU'LL HAVE AFTER STEP 4

Directory Structure

```
fluoride-adsorption-hybrid/
```

```
data/
    doe_matrix.csv              ← 48 experimental runs
    dataset_simulated.csv       ← Complete dataset with responses

src/
    doe_design.py               ← DoE matrix generation
    simulate_adsorption.py      ← Physics-based simulation

notebooks/
    01_data_exploration.py      ← Data visualization
    02_langmuir_fitting.py      ← (Phase 2 - coming)
    03_ml_training.py           ← (Phase 4 - coming)
    04_hybrid_analysis.py       ← (Phase 5 - coming)

output/
    01_data_exploration.png     ← Visualization

models/
    langmuir_ssl.pkl            ← (Phase 2 - coming)
    ml_model_rf.pkl             ← (Phase 4 - coming)
    hybrid_pipeline.pkl         ← (Phase 5 - coming)

reports/
    (Final reports and figures)

requirements.txt                ← Python dependencies
.gitignore                      ← Git ignore file
README.md                       ← Project documentation
```

Data Overview

`dataset_simulated.csv` (48 rows × 7 columns)

Run	pH	C0_mg_L	Time_min	Temp_C	Flow_L_min	Efficiency_percent	Capacity_mg_g
1	3.0	1.0	10	20	0.5	42.3	0.42
2	6.0	5.5	65	30	1.25	89.2	4.90
...	...	...	...	...	...	...	...

KEY DECISION: SIMULATED VS REAL DATA

Option A: Continue with Simulated Data RECOMMENDED

Advantages: - **Fast iteration:** Generate data in minutes - **Perfect reproducibility:** Same seed → same results - **Theory validation:** Confirms hybrid model approach works - **No lab access required:** Works anywhere - **Budget-friendly:** No experimental costs

Disadvantages: - Not real lab data - May miss unexpected phenomena

Best use: Validate methodology, develop code, tune hyperparameters

Decision: START HERE (Simulated data) → **EXTEND TO** real data if needed

---

Option B: Use Real Experimental Data

Advantages: - Realistic results - Publishable quality - Handles real variability

Disadvantages: - Requires lab access or literature data - Slower (weeks to months) - Higher cost

Best use: After validating approach with simulated data

Decision: Use simulated → Validate → Then run real experiments

---

IMMEDIATE ACTION ITEMS

Quick Start (Next 30 minutes)

```
# 1. Navigate to project
cd fluoride-adsorption-hybrid

# 2. Activate environment
source venv/bin/activate

# 3. Run DoE generation
python src/doe_design.py

# 4. Run simulation
python src/simulate_adsorption.py

# 5. Check output
ls -lh data/*.csv
python -c "import pandas as pd; print(pd.read_csv('data/dataset_simulated.csv').shape)"
```

Expected Completion Time

- Total: ~**30-45 minutes**
- DoE generation: **5 minutes**
- Simulation: **2 minutes** (all 48 runs)
- Validation: **5 minutes**
- Exploration plots: **10 minutes**

TROUBLESHOOTING & VALIDATION

Common Issues & Fixes

Issue 1: `ModuleNotFoundError: No module named 'pyDOE2'` **Fix:**

```
pip install pyDOE2
```

Issue 2: `FileNotFoundError: 'data/doe_matrix.csv'` **Fix:** Make sure you ran `doe_design.py` first:

```
python src/doe_design.py
```

Issue 3: Simulation produces negative efficiencies **Fix:** Check the simulation parameters. The code clips values to [0, 100], but if you see warnings, the noise might be too high. Default is ±5%, which is correct.

Issue 4: Data doesn't show pH bell curve **Fix:** Increase sample size by increasing center points in CCD:

```python
# In doe_design.py
design = ccdesign(n=5, center=8, face='face')  # Increase from 6 to 8
```

Validation Checklist

After running all scripts, verify:

- ☐ `data/doe_matrix.csv` exists and has 48 rows
- ☐ `data/dataset_simulated.csv` exists and has 48 rows
- ☐ Efficiency values are between 0-100%
- ☐ Capacity values are positive (0.1 to ~10 mg/g)
- ☐ pH shows bell curve (peak at 6.5)
- ☐ Time shows saturation curve
- ☐ Temperature shows positive effect
- ☐ Flow rate shows negative effect
- ☐ Output plot generated at `output/01_data_exploration.png`

NEXT PHASE: LANGMUIR FITTING (Phase 2)

Once data is ready, the next phase will:

1. **Fit Langmuir isotherm** (SSL model)
   - Extract qmax and KL values
   - Calculate R^2, RMSE, residuals
2. **Test Dual-Site Langmuir** (DSL)
   - Heterogeneous surface modeling
   - Compare with SSL
3. **Residual analysis**

- Plot residuals vs pH, time, temperature
- Identify systematic patterns

4. **ML feature engineering**
   - Create ~10 physics-informed features
   - Prepare for model training

KEY METRICS TO TRACK

Performance Indicators

Metric	Phase 2 Target	Phase 5 Target
Chemical Model R^2	0.85	—
ML Model R^2	—	0.95
Hybrid Model R^2	—	0.92
RMSE (Chemical)	1.2 mg/g	—
RMSE (Hybrid)	—	0.8 mg/g
Improvement	—	15-30% vs Chemical

COMMAND REFERENCE

Setup

```bash
# Create and activate environment
python -m venv venv
source venv/bin/activate  # or: venv\Scripts\activate

# Install dependencies
pip install pyDOE2 numpy pandas scipy scikit-learn matplotlib seaborn jupyter

# Save requirements
pip freeze > requirements.txt
```

Execution

```bash
# Generate DoE
python src/doe_design.py

# Run simulation
python src/simulate_adsorption.py

# Explore data
python notebooks/01_data_exploration.py

# View results
```

```
cat data/doe_matrix.csv | head -5
cat data/dataset_simulated.csv | head -5
```

Verification

```
# Check files exist
ls -lh data/*.csv output/*.png

# Summary statistics
python -c "
import pandas as pd
df = pd.read_csv('data/dataset_simulated.csv')
print(df.describe())
"
```

DOCUMENTATION REFERENCES

- **Phase 1 Report:** `phase1_research_report.md` (40 pages of literature)
- **DoE Design:** Face-Centered CCD with 5 factors, 48 runs
- **Simulation:** Physics-based model with 6 mechanisms
- **Validation:** Checked against 30+ literature sources

SUCCESS CRITERIA

You've completed Phase 1 Execution when:

1. `data/doe_matrix.csv` created (48 runs, 5 factors)
2. `data/dataset_simulated.csv` created (48 runs, 2 responses)
3. Data validated against literature trends
4. Visualization generated showing all effects
5. Summary statistics reviewed and documented
6. Ready to proceed to Phase 2 (Langmuir Fitting)

QUICK START COMMAND

Copy and paste this to run everything:

```
cd fluoride-adsorption-hybrid
source venv/bin/activate
python src/doe_design.py && python src/simulate_adsorption.py && python notebooks/01_data_explo
echo "  Phase 1 Execution Complete!"
```

NEED HELP?

If you encounter issues:

1. **Check the troubleshooting section** above
2. **Verify all dependencies:** `pip list | grep -E "numpy|pandas|pyDOE"`
3. **Check file permissions:** `ls -la data/`
4. **Review the scripts:** The code is well-commented with inline explanations

---

FINAL NOTES

- **Estimated Time for Phase 1.1 + 1.2:** 3-4 hours total
- **Estimated Time for Phase 2-8:** 20-25 hours total
- **Total Project Timeline:** 3-4 weeks recommended
- **Skills Gained:** DoE, physics-based simulation, Langmuir modeling, ML, hybrid systems

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Next Action: Run the scripts and share the data summary with Phase 2 starting next!

Ready to execute Phase 1 Execution? Start now!

Document prepared: May 3, 2026
Status: Ready for execution
Last updated: Phase 1 Research Complete